R for Beginners: From Zero to Data-Driven Analysis
Solidarity-based course supporting Sudanese students and researchers
Instructor: Dr. Basem Aljoumani
Course Coordinator: Dr. Sarra Ahmed Mohamed Saad

Mini Assessment – Lessons 1–6
Due Date: Friday, May 29, 2026

This assessment is designed to review the concepts learned in Lessons 1–6, including variables, vectors, matrices, data frames, logical conditions, and decision-making using if statements.

Imagine that you are part of a small environmental monitoring team collecting observations from different cities. Use your R skills to explore and analyze the data below.

Instructions:

- **Write your R code clearly.**
- **Run your code and check your results.**
- **Save your script with your name.**
- **You may use your lesson materials and notes**

The following environmental data were collected from five cities.

```
city <- c("Berlin", "Paris", "Rome", "Madrid", "Tunis")

temperature <- c(18, 22, 27, 30, 35)

rain <- c(5, 2, 0, 1, 0)

soil_moisture <- c(0.25, 0.22, 0.15, 0.12, 0.08)
```

Part 1: Variables and Basic Calculations

5 points

Q1. Create two variables:

```
x <- 12
y <- 4
```

Calculate:

- x + y
- x * y
- x / y

- x^2

Part 2: Data Types and Vectors

5 points

Q2. Check the class of these objects:

city
temperature
soil_moisture

Q3. Give names to the temperature vector using the city names.

Part 3: Vector Filtering and Logical Thinking

6 points

Q4. Write R code to answer:

- Which cities have temperature greater than 25?
- Which soil moisture values are less than 0.20?
- Which cities have no rain?

Part 3B: Matrix Challenge (2 points)

A <- matrix(1:12, nrow = 3)

- What are the dimensions of A?
- Show the second row.
- Show the first column.

Part 4: Data Frames

6 points

Q5. Create a data frame called env_data **using the four vectors, include the variables as columns.**

Q6. Show:

- The first rows using head()
- The structure using str()
- The number of rows using nrow()

Part 5: Filtering a Data Frame

4 points

Q7. From env_data**, select cities where:**

- temperature is greater than 25
- rain is equal to 0
- temperature is greater than 25 and soil moisture is less than 0.20

Part 6: Simple Decision Rule with if / else

4 points

Q8. Use this value:
sm <- 0.12

Write an if…else statement:
If sm < 0.15, print:

"Soil is dry"

Otherwise print:

"Soil moisture is acceptable"

Bonus Question

Optional: 3 points

Create a logical column in env_data called:

water_stress

Apply the following rule:

TRUE → when soil moisture is less than 0.15
FALSE → otherwiseThe value should be:

Good luck